

All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization

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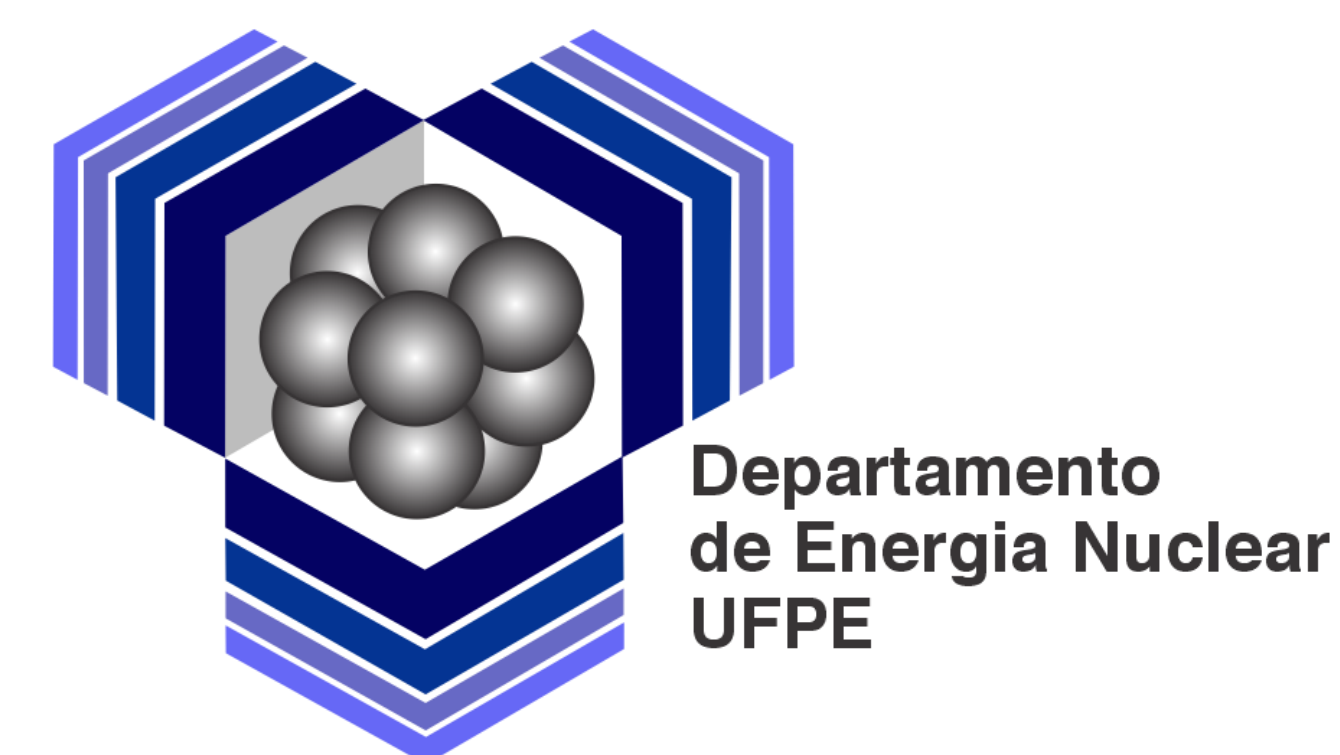
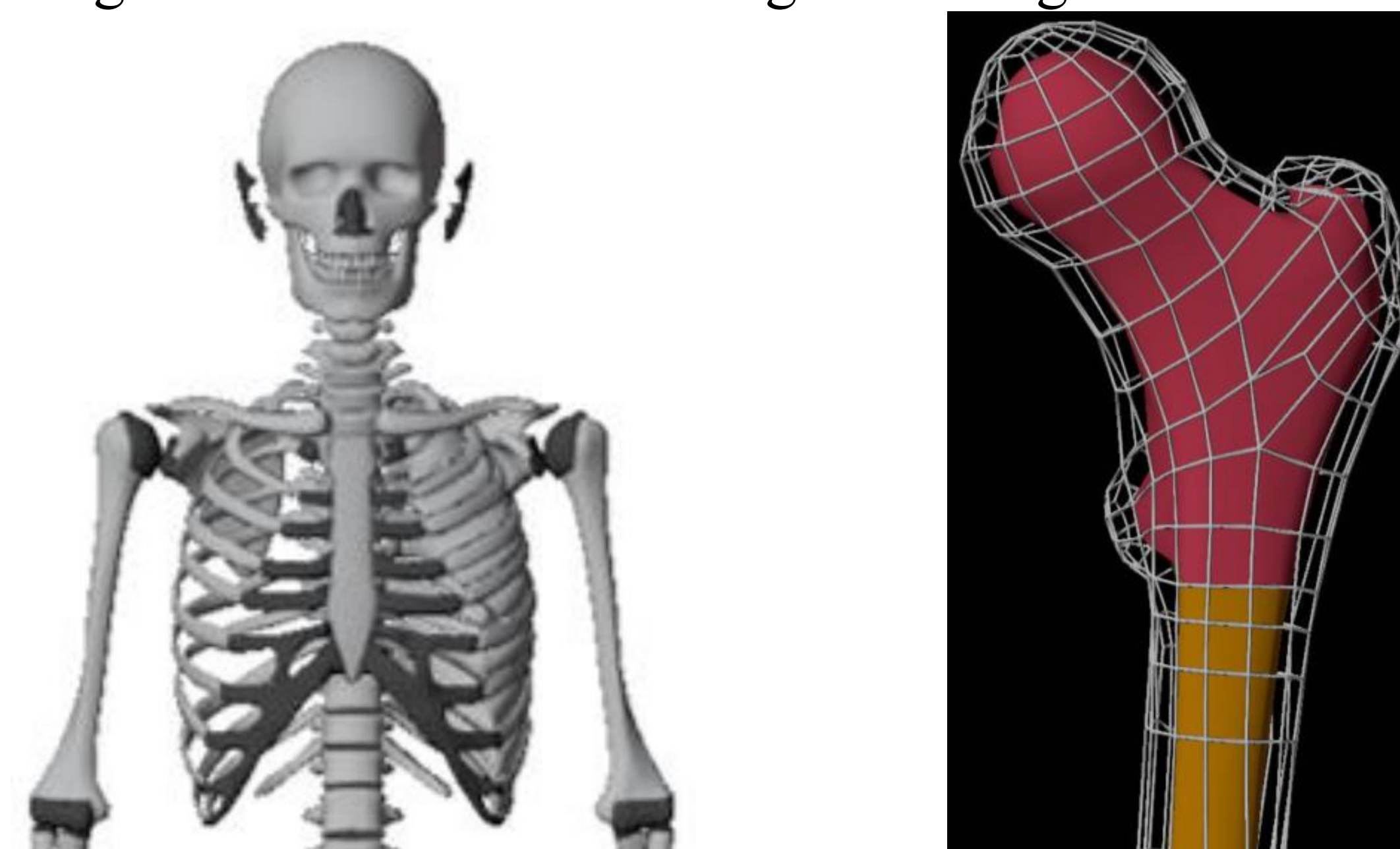
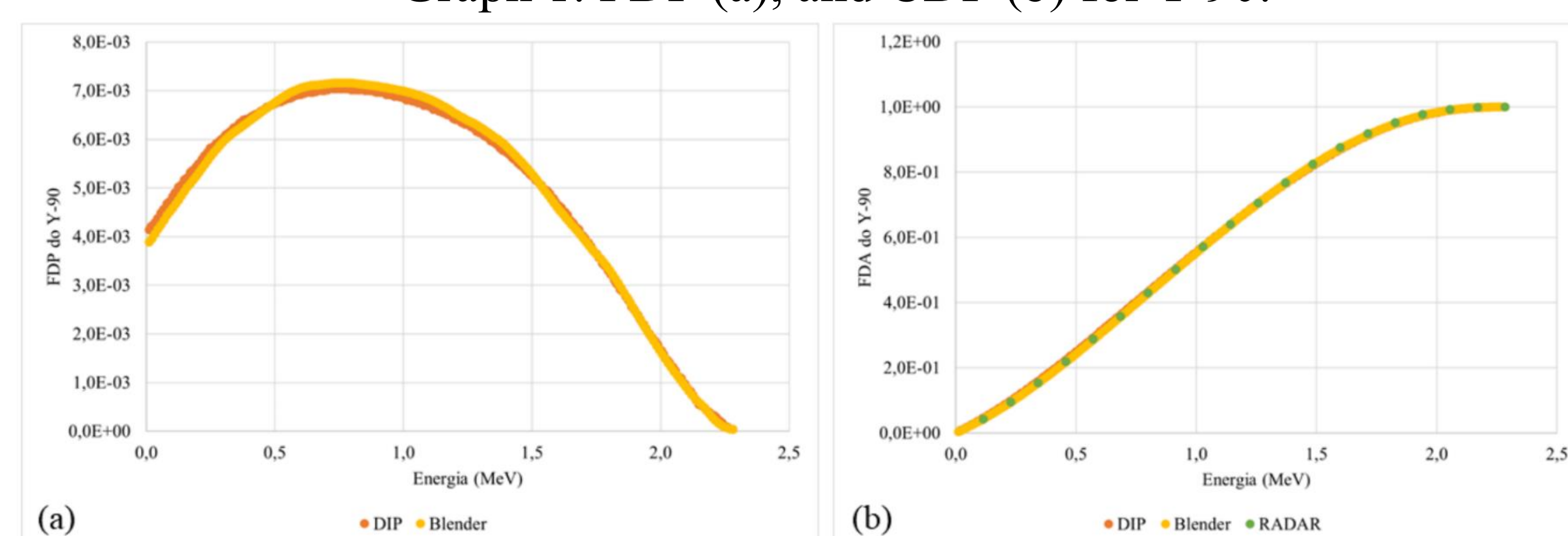


Figure 1. MARTIN'S Cartilages and Segmented Bones.



Source: the authors, 2025.

Graph 1. PDF (a), and CDF (b) for Y-90.



Source: the authors, 2025.

INTRODUCTION

The anthropomorphic phantom MARTIN (Male Adult with Macro Circulation and Lymphatic Vessels Phantom), created in 2018 by Andrade, is a Boundary Representation Model (BREP) built with polygonal meshes that includes a network of blood and lymphatic vessels [1]. This work describes the methodology and results of the MARTIN phantom update. Modifications were made to increase the model's realism by incorporating additional structures such as cancellous bones and a medullary cavity, in addition to correcting posture, adjusting muscles, including cartilage, and repositioning the lungs [2]. To demonstrate the applicability of MARTIN in internal dosimetry, we implemented an Exposure Computational Model (ECM). This model is based on the MASH-RADIOEMBOLIZATION/EGSnrc framework, which uses widely recognized dosimetry methodologies to simulate radioembolization procedures, employing the Monte Carlo code EGSnrc [3].

MATERIALS AND METHODS

- Blender v. 2.83.20: Used for modifications to MARTIN and the PHAMA code implementation environment;
- PHAMA v.1.1.0 (in-house): Blender add-on used for modifications to MARTIN and modeling of the Y-90 source.
- OrtoGOnBlender v. 2021.12.25: add-on with the toolkit used for voxel reconstruction of abdominal tomography data for liver editing;
- Digital Image Processing (DIP): In-house software used for MARTIN voxelization, creation of cumulative distribution function (CDF) and probability density function (PDF) from source and input files for MCE

The modifications to MARTIN were made using 3D modeling techniques, using Blender and PHAMA. In the latter, we applied non-destructive modeling methods. For the liver, we used OrtoGOnBlender to reconstruct a patient's CT scan in the 3D environment, allowing us to enhance the organ's proportions without losing its anatomical fidelity. The Y-90 source was modeled in PHAMA, where the software tools generated contour surfaces and an output file with relative heights, which correspond to the probability of beta particle emission as a function of its energy. Finally, DIP was used to voxelize MARTIN and generate the input files for MCE.

RESULTS

The inclusion of new organs and tissues in the MARTIN phantom increased the number of VOIs (Volumes of Interest) from 104 to 131. The main additions were the segmentation of the skeleton in the mesh phase and the inclusion of cartilage, as illustrated in Figure 1. For coupling to EGSnrc, the phantom was voxelized using DIP software. This process generated a data stack with 443 columns, 272 rows, and 1,467 slices, using voxels with 1.2 mm edges.

Analysis of the PDF and CDF obtained by PHAMA/DIP shows high consistency with the Radiation Dose Assessment Resource (RADAR) reference data, as shown in Graph 1.

The results expressed as dose per cumulative activity ($D/\tilde{A}$) were below 5% for all VOIs as shown in Table 1.

Table 1. Dosimetric results.

ID	NAME	D/Ac (mGy/MBq.s)	CV (%)
3	Bladder Wall	6.0867E-10	3.60
4	Brain	4.6872E-11	3.05
6	Colon Wall	3.7899E-09	0.51
7	Breast Glands	3.5627E-09	3.59
9	Liver	8.3844E-05	0.00
10	Lungs	5.0406E-09	0.25
17	Skin	9.3617E-10	0.36
19	Stomach Wall	4.4187E-08	0.23
20	Salivary Glands	2.7004E-10	4.61
68	Total RBM	4.6952E-09	0.23
89	Total BSC	2.4998E-09	0.36

Source: the authors, 2025.

CONCLUSIONS

The low coefficient of variation for the VOIs proves the effectiveness and accuracy of the simulations performed with 1E8 particles.

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